

# Stage 11 Revision 47 Provenance, PMF, Kinetic Cross-Fit, and Typed-DAG Consistency

mdstats

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## Purpose

This specification prevents thermodynamic-source ambiguity, mandatory cross-check over-gating, PMF estimator conflation, in-sample kinetic validation, zero-event edge loss, candidate-freeze ambiguity, source mixing, and weak dependency-edge semantics.

## Thermodynamic provenance and optional verification

Every thermodynamic result must include `ThermodynamicResultProvenance`, including the estimator, source bundle, source records/channels, ensemble and target measure, state semantics, coordinate measure/Jacobian, estimation partition, sampling, uncertainty, and assumptions. A result whose own gates pass may be reported as `source_qualified_unverified`. Cross-checking is optional unless the caller explicitly requests a `cross_validated` product.

Verification status is one of `not_requested`, `unavailable`, `insufficient`, `agreed`, `partially_agreed`, or `disagreed`. Disagreement blocks only a combined consensus and retains both source estimates.

## PMF estimator split

- `PMF_DENSITY` produces `DensityPmfCertificate` without requiring force data.
- `PMF_FORCE` produces `ForcePmfCertificate` from E3B admissible mean-force evidence.
- `PMF_CROSSCHECK` is optional and compares compatible density and force PMFs.

## Cross-fitting and rates

**EvidenceCrossfitPartition** separates thermodynamic estimation from optional verification. **KineticCrossfitPartition** separately owns **kinetic\_model\_selection**, **kinetic\_model\_fit**, and **kinetic\_model\_validation** complete episodes. F0 uses fit evidence; E9B uses untouched validation evidence. Crossing episodes are censored under a signed boundary policy.

**RateCandidateEdgeUniverse** is declared before rate fitting. It retains candidates with valid at-risk exposure even when zero events are observed. Candidate edges and observed event edges are distinct.

## Candidate freeze and source identity

E2 emits per-realization candidate catalogs. GR4 alone emits the **FrozenCandidateCatalog**. All source-derived records carry one **SourceTrajectoryBundleIdentity**; controls and coordinates from different bundles may not be combined.

## Typed DAG

`stage11_dependency_graph.json` schema version 2 uses typed edges:

```
source_identity_requires
execution_requires
promotion_requires
conditional_requires
optional_enrichment
optional_verification
supersedes
replay_triggers
```

Tests must verify source identity, required-edge acyclicity, PMF split, optional thermodynamic verification, kinetic fit/validation separation, zero-event candidate-edge support, GR4 freeze ownership, persistent model-generation records, and product-specific E8b gates.

## Persistent model-generation contract

**StateModelGeneration** records the state catalog, parent generation, segmentation policy, rate-candidate policy, and all partition signatures. **GenerationReplayPlan** lists the bounded stages to replay after gating or semi-Markov revision. **GenerationTerminationCertificate** records accepted, unsupported, cycle-prevented, maximum-generation, or sampling-insufficient termination.

## E8b product separation

Geometry, thermodynamic, corridor, path, transition-state, event-network, and kinetic comparisons are distinct products. THERMO4A is optional verification for basin thermodynamics and cannot gate geometry or occupancy comparison. E6b gates path comparison; THERMO3B gates transition-state comparison.

## Grid and correspondence policy

`stage11_grid_stopping_v1` and `stage11_feature_correspondence_v1` are versioned initial presets. Exact values are serialized. A changed tolerance or cost definition requires a new identifier and cannot silently change prior certificates.